

Chapter 25 Back Matter Check

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25 Supersymmetry Algebras

Problems

1. Find a set of $2 \cdot 2$ matrices that form a graded Lie algebra containing fermionic as well as bosonic generators.
2. Following the approach of Haag, Lopuszanski, and Sohnius, derive the form of the most general symmetry superalgebra in $2 + 1$ spacetime dimensions. (Hint: With the generators of the Lorentz group in $2 + 1$ spacetime dimensions labelled

$$A_1 = -iJ_{10}, \quad A_2 = -iJ_{20}, \quad A_3 = J_{12},$$

the commutation relations of the Poincaré algebra are

$$[A_i, A_j] = i \sum_k \epsilon_{ijk} A_k,$$

so the representations of the homogeneous Lorentz group in $2 + 1$ spacetime dimensions are labelled with a single positive integer or half-integer A .) Assume that the conditions for the Coleman–Mandula theorem are satisfied here.

3. Suppose that there were no massless particles with helicity greater than $+3/2$ or less than $-3/2$. Find the most general massless particle multiplets for $N = 6$ extended supersymmetry and (using CPT symmetry) $N = 5$ extended supersymmetry. What does the comparison between the particle content you find suggest about these two extended supersymmetries?
4. What are the possible parities of the particles in the short supermultiplets of extended $N = 2$ supersymmetry?

References

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